Media Mix Modelling

Measuring marketing effectiveness for a fashion retail client

DATA

117 weeks of sales, marketing & macro data

METHOD

OLS regression with adstock & log transforms

OBJECTIVE

Identify channel ROI & optimise budget allocation

Data & Processing

DATA

Weekly file: 117 rows - sales, marketing spend, competition, weather, brand awareness

Monthly file: 36 rows - unemployment, GDP, consumer confidence, tourists

Merged via year-month key. Monthly values repeat for each week within the month.

TRANSFORMATIONS

Log transforms applied to sales, marketing spend, economic & competition variables

Why: comparable scales + coefficients become elasticities (% change interpretation)

Marketing channels: log(x+1) to handle zero-spend weeks

ADSTOCK

TV (decay 0.5): longer carryover. Correlation improved 0.05 → 0.20

Press (decay 0.3): shorter-lived effect, print fades faster

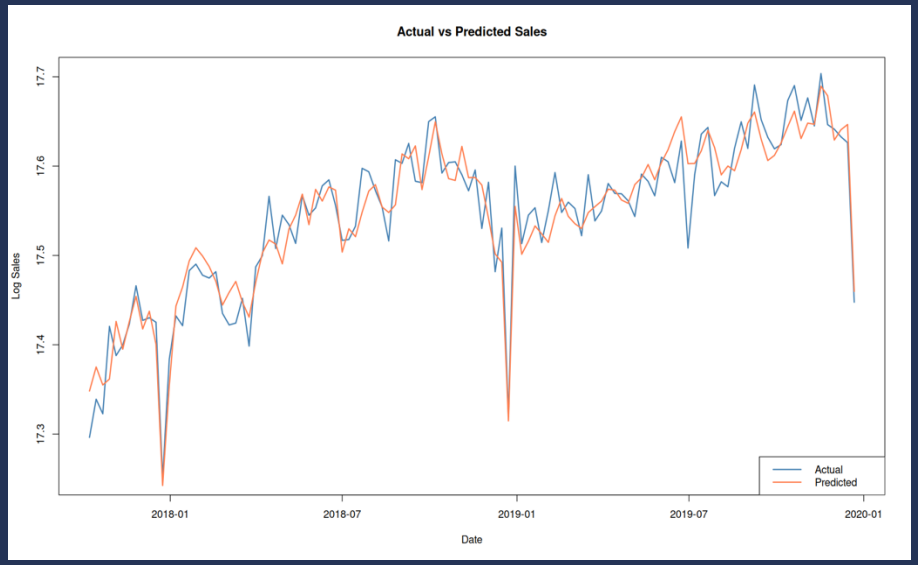
Not applied to: online (direct response), banners (immediate), radio (insufficient data)

Not transformed: Public holidays & Christmas dummy (binary), brand knowledge & competitor recognition (proportions 0–1), weather & precipitation indices. These can go into the model in raw form.

Final Model Structure

VARIABLE	GROUP	COEFFICIENT	P-VALUE	VIF	SIGN
Lagged Sales	Structural	+0.158	0.006 **	4.08	✓ Expected
Online	Marketing	+0.014	0.008 **	1.38	✓ Expected
Press (adstock)	Marketing	-0.003	0.011 *	1.41	⚠ Reverse causality
Economy Index	Economic	+0.650	<0.001 ***	2.36	✓ Expected
Unemployment	Economic	-0.476	<0.001 ***	8.25	✓ Expected
Consumer Confidence	Economic	+0.650	<0.001 ***	3.01	✓ Expected
Tourists	Economic	-0.080	0.002 **	3.55	✓ Locals crowded out
Competitor Recognition	Competition	-0.178	0.005 **	1.26	✓ Expected
Weather Index	External	-0.001	0.008 **	2.37	✓ Expected
Public Holidays	External	-0.021	<0.001 ***	2.11	✓ Expected
Christmas Dummy	External	-0.113	<0.001 ***	2.18	✓ Expected

Model Validation



Model tracks actual sales closely across the full two-year period

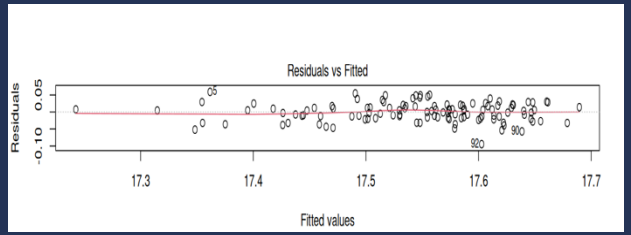
DIAGNOSTIC CHECKS

Adjusted R^2
0.897
✓ **90% of sales variation explained**

Durbin-Watson
1.91
✓ **No autocorrelation**

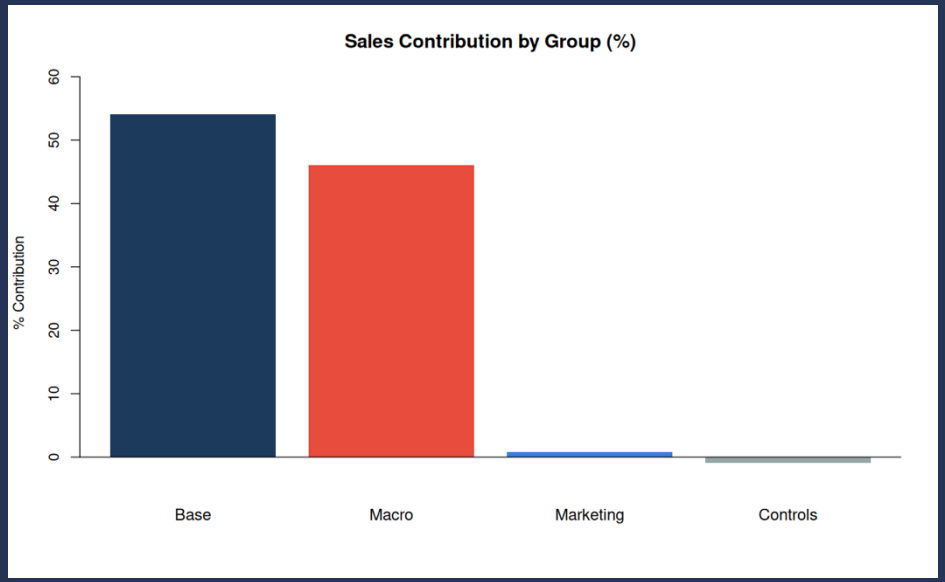
Shapiro-Wilk
p = 0.51
✓ **Residuals normally distributed**

VIF
All < 10
✓ **No multicollinearity issues**



What Drives Sales?

Macroeconomic conditions and structural brand strength explain over 99% of sales - marketing contributes < 1%



Base
54.1% Structural brand sales & weekly momentum

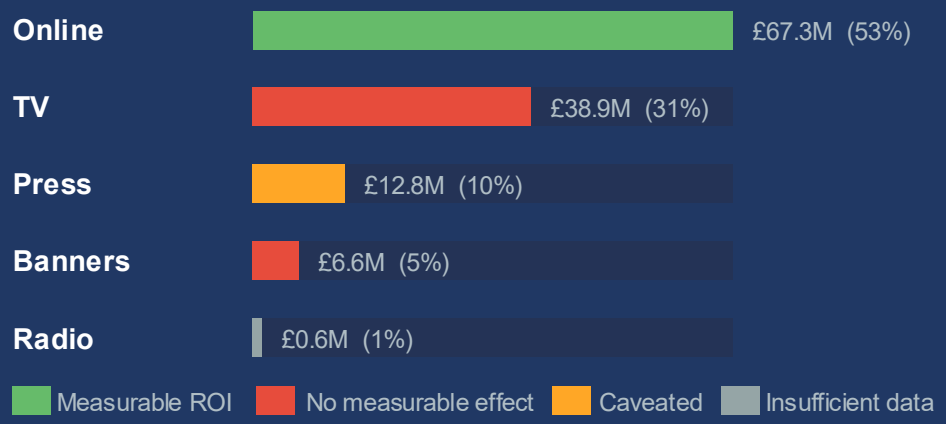
Macro
46.0% Economy, confidence, unemployment, tourists

Marketing
0.8% Online (positive), press (caveated)

Controls
-0.9% Weather, holidays, competition

Marketing Effectiveness & Recommendations

BUDGET ALLOCATION



Online ROI: £0.98 per £1 spent

Direct sales return only - true ROI likely higher with brand awareness value

CHANNEL VERDICT

Online
Maintain or increase. Only channel with measurable return

TV
No measurable impact despite 31% of budget. Question allocation, but may have long-term brand effects

Press
Negative coefficient likely reflects reverse causality. Review campaign timing

Banners
No measurable effect on sales

Radio
Only active 6/117 weeks - insufficient data to assess. Needs a sustained campaign to measure effect